

REPORT

ARTIFICIAL INTELLIGENCE (CSA1717)

ASSESSMENT TOOL 3 – RESOLUTION ALGORITHM IMPLEMENTATION

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1. INTRODUCTION

Resolution is an important inference technique used in Artificial Intelligence and automated reasoning. It is used to determine whether a logical statement follows from a given knowledge base.

The Resolution Algorithm works by converting logical statements into Conjunctive Normal Form (CNF), negating the required goal, and repeatedly resolving clauses containing complementary literals. If the process produces the empty clause $\square$, the original goal is logically proved.

In this assessment, the Resolution Algorithm is implemented using Python for five different logical reasoning problems: rain and wet ground, student assignment submission, library membership, placement eligibility, and access control.

2. RESOLUTION ALGORITHM

The general procedure of Resolution Refutation is:

1. Represent the knowledge base using logical propositions.
2. Convert implications into CNF.
3. Represent the resulting statements as clauses.
4. Negate the goal.
5. Add the negated goal to the knowledge base.
6. Select two clauses containing complementary literals.
7. Resolve the complementary literals.
8. Generate a new clause.
9. Continue the process until:
 - the empty clause $\square$ is obtained, or
 - no new clause can be generated.
10. If the empty clause is obtained, the goal is proved.

Resolution Principle

For two clauses:

$A \vee B$

and

$\neg B \vee C$

Resolving complementary literals **B** and **$\neg B$** gives:

$A \vee C$

When a contradiction is obtained:

P

and

$\neg P$

the result is:

$\square$

The empty clause proves the original goal.

3. QUESTION 1 – RAIN AND WET GROUND

3.1 Aim

To implement the Resolution Algorithm and prove that the ground is wet based on the given knowledge base.

3.2 Problem Statement

The knowledge base contains:

1. If it rains, then the ground becomes wet.
2. It is raining.

The goal is to prove that the ground is wet.

3.3 Propositional Symbols

- R = It is raining.
- W = Ground is wet.

3.4 Knowledge Base

Rule:

$$R \rightarrow W$$

Fact:

$$R$$

Goal:

$$W$$

3.5 CNF Conversion

Using:

$$R \rightarrow W \equiv \neg R \vee W$$

Therefore:

$$C1 = \neg R \vee W$$

$$C2 = R$$

3.6 Negated Goal

Goal:

$$W$$

Negated goal:

$$C3 = \neg W$$

3.7 Resolution

Resolve C1 and C2:

$$\neg R \vee W$$

$$R$$

Therefore:

$$C4 = W$$

Now resolve C4 and C3:

$$W$$

$$\neg W$$

Therefore:

□

3.8 Result

The empty clause is obtained.

Ground is Wet – PROVED.

4. QUESTION 2 – STUDENT ASSIGNMENT SUBMISSION

4.1 Aim

To implement the Resolution Algorithm and prove that Rahul receives internal marks after submitting his assignment.

4.2 Problem Statement

The knowledge base contains:

1. If a student submits the assignment, the student receives internal marks.
2. Rahul submitted the assignment.

The goal is to prove that Rahul receives internal marks.

4.3 Propositional Symbols

- S = Rahul submitted the assignment.
- M = Rahul receives internal marks.

4.4 Knowledge Base

Rule:

$$S \rightarrow M$$

Fact:

$$S$$

Goal:

M

4.5 CNF Conversion

$$S \rightarrow M \equiv \neg S \vee M$$

Therefore:

$$C1 = \neg S \vee M$$

$$C2 = S$$

4.6 Negated Goal

$$C3 = \neg M$$

4.7 Resolution

Resolve:

$$C1: \neg S \vee M$$

with:

$$C2: S$$

Therefore:

$$C4 = M$$

Now resolve:

$$C4: M$$

with:

$$C3: \neg M$$

Therefore:

□

4.8 Result

The empty clause is obtained.

Rahul receives internal marks – PROVED.

5. QUESTION 3 – LIBRARY MEMBERSHIP

5.1 Aim

To implement the Resolution Algorithm and prove that Priya can borrow books because she is a library member.

5.2 Problem Statement

The knowledge base contains:

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

The goal is to prove that Priya can borrow books.

5.3 Propositional Symbols

- L = Priya is a library member.
- B = Priya can borrow books.

5.4 Knowledge Base

Rule:

$$L \rightarrow B$$

Fact:

$$L$$

Goal:

$$B$$

5.5 CNF Conversion

$$L \rightarrow B \equiv \neg L \vee B$$

Therefore:

$$C1 = \neg L \vee B$$

$$C2 = L$$

5.6 Negated Goal

$$C3 = \neg B$$

5.7 Resolution

Resolve C1 and C2:

$$\neg L \vee B$$

$$L$$

Therefore:

$$C4 = B$$

Resolve C4 and C3:

$$B$$

$$\neg B$$

Therefore:

□

5.8 Result

The empty clause is obtained.

Priya can borrow books – PROVED.

6. QUESTION 4 – PLACEMENT ELIGIBILITY

6.1 Aim

To implement the Resolution Algorithm and prove that Arun is eligible for placement after clearing the aptitude test.

6.2 Problem Statement

The knowledge base contains:

1. If a student clears the aptitude test, the student is eligible for placement.
2. Arun cleared the aptitude test.

The goal is to prove that Arun is eligible for placement.

6.3 Propositional Symbols

- A = Arun cleared the aptitude test.
- P = Arun is eligible for placement.

6.4 Knowledge Base

Rule:

$$A \rightarrow P$$

Fact:

A

Goal:

P

6.5 CNF Conversion

$$A \rightarrow P \equiv \neg A \vee P$$

Therefore:

$$C1 = \neg A \vee P$$

$$C2 = A$$

6.6 Negated Goal

$$C3 = \neg P$$

6.7 Resolution

Resolve:

$$\neg A \vee P$$

with:

$$A$$

Therefore:

$$C4 = P$$

Now resolve:

$$P$$

with:

$$\neg P$$

Therefore:

□

6.8 Result

The empty clause is obtained.

Arun is eligible for placement – PROVED.

7. QUESTION 5 – ACCESS CONTROL SYSTEM

7.1 Aim

To implement the Resolution Algorithm and prove that the user is granted access after entering the correct password.

7.2 Problem Statement

The access control system contains:

1. If a user enters the correct password, the user is authenticated.
2. If a user is authenticated, the user is granted access.
3. The user entered the correct password.

The goal is to prove that the user is granted access.

7.3 Propositional Symbols

- P = User entered the correct password.
- A = User is authenticated.

- G = User is granted access.

7.4 Knowledge Base

Rule 1:

$$P \rightarrow A$$

Rule 2:

$$A \rightarrow G$$

Fact:

$$P$$

Goal:

$$G$$

7.5 CNF Conversion

Rule 1:

$$P \rightarrow A$$

Therefore:

$$C1 = \neg P \vee A$$

Rule 2:

$$A \rightarrow G$$

Therefore:

$$C2 = \neg A \vee G$$

Fact:

$$C3 = P$$

7.6 Negated Goal

Goal:

$$G$$

Negated goal:

$$C4 = \neg G$$

7.7 Resolution

Step 1

Resolve C1 and C3:

$$\neg P \vee A$$

P

Therefore:

C5 = A

Step 2

Resolve C2 and C5:

$\neg A \vee G$

A

Therefore:

C6 = G

Step 3

Resolve C6 and C4:

G

$\neg G$

Therefore:

□

7.8 Logical Flow

Correct Password

↓

Authenticated

↓

Granted Access

↓

Negated Goal $\neg G$

↓

Empty Clause □

7.9 Result

The empty clause is obtained.

The user is granted access – PROVED.

9. REFERENCES

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